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REGULAR RESEARCH PAPER

A novel ensemble learning-based model for APT attack detection

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Abstract

Improving the effectiveness of APT attack detection models is one of the most critical and essential tasks today. Following this trend, this paper proposes a new model called ACDF-mLSTM to address two primary challenges currently faced by research in this field: (i) data imbalance and (ii) information aggregation and feature extraction. Specifically, to solve the data imbalance problem, the paper proposes a novel data generation method named ACDF. This method leverages advanced and sophisticated techniques to focus on identifying crucial points in sequential data and analyzing context by considering preceding and succeeding data points. Subsequently, a Diffusion Model is applied to generate synthetic APT attack data, built upon the principle of gradual diffusion. With this approach, the ACDF model can generate more meaningful and realistic data. Next, to address the task of information aggregation and feature extraction, the paper proposes a new deep learning model named mLSTM, based on the optimization of Long Short-Term Memory (LSTM). Thus, the mLSTM model performs two main tasks: (i) extracting information from network flows within traffic and (ii) aggregating and highlighting important information before it enters the classification model. In the experimental section, the paper evaluates the ACDF-mLSTM model for the first time across various scenarios and datasets to demonstrate its effectiveness and adaptability. The evaluation results show that the ACDF-mLSTM model outperformed most other methods by an average of 2 to 12% across all metrics and on all experimental datasets.

Keywords APT attack detection · Ensemble learning model · Advanced deep learning models · Rebalancing data

1 Introduction

1.1 Overview

Advanced Persistent Threat (APT) attacks represent one of the greatest challenges to modern cybersecurity due to their sophisticated, persistent nature and their ability to evade traditional detection systems [41]. Unlike conventional cyber-attacks, APTs are typically meticulously planned by organized hacking groups to infiltrate critical systems such as government agencies, financial institutions, and essential infrastructure. The primary objective of these attacks is not only to steal data but also to disrupt operations and even manipulate systems from within. Using advanced techniques such as zero-day exploits, lateral movement, and encrypted command and control (C2) channels, attackers can maintain a difficult-to-detect presence within a system for extended periods. [42], [43] This makes it challenging for traditional rule-based Intrusion Detection Systems (IDS) to identify APT activities.

To address the task of early detection and warning of APT attacks, models based on the analysis of flows in network traffic have recently emerged as an effective approach, owing to their ability to identify abnormal behaviors in a detailed and accurate manner [1–3]. The trend in these studies is to focus on several main stages, such as information extraction, and the aggregation and highlighting of important information and features. For the feature extraction stage of APTs based on network traffic, studies often use deep learning techniques [4], deep graph networks [5–9], and other methods. For the stage of aggregating and highlighting important information, models like Attention mechanisms, inference, or Transformers [10, 11] are often preferred. Based on such stages and lifecycles, ensemble learning models are commonly constructed to combine the advantages of individual methods

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into a complete solution [12–16]. However, upon researching and evaluating recent ensemble learning techniques, we have identified several issues that need to be improved to enhance the accuracy of APT attack detection and reduce the rate of false alarms and misclassifications in these models. These include:

- The Problem of Data Imbalance: Generally, some recently proposed models have addressed the challenge of data balancing in imbalanced datasets. Several methods have been applied, such as the Synthetic Minority Over-sampling Technique (SMOTE) [17], Dropout [18], VAE [19], conditional Generative Adversarial Networks (cGAN) [20], and Generative Adversarial Networks (GAN) [21]. However, these methods typically choose to generate feature vectors for the minority classes after the information aggregation and extraction stage. A notable drawback of this approach is that the generated data is prone to noise and may contain misleading or meaningless information, leading to suboptimal performance. This is because the feature vectors generated after the aggregation and extraction process have already been transformed and processed through a series of complex models. Consequently, these feature vectors differ significantly from the original feature values. Therefore, if these vectors then go through a data balancing stage, the resulting vectors will not hold much meaning or value to improve classification.
- The Problem of Feature Aggregation and Extraction: Some recent studies have indicated that although deep learning models, deep graph networks, or Transformers have been proposed, they often process data sequentially without considering the special relationships within the data. This creates certain problems, such as failing to filter and focus on important data, which leads to information loss, a lack of focus on the problem at hand, and the introduction of noise into the learned information. The inability to self-adjust to datasets with varying numbers of attack samples and the failure to capture the complex relationships of sophisticated attack patterns limit the model's applicability to diverse datasets or new attack data.

1.2 Problem resolution

To address the two aforementioned problems, this paper proposes an Ensemble Learning model comprising two main components: data generation using the ACDF model, and APT feature aggregation and extraction based on the mLSTM model. A final classification component is also included. The ACDF-mLSTM Ensemble Learning model addresses the two problems faced by other models as follows:

First, regarding the data generation problem As previously mentioned, the ACDF-mLSTM model differs from traditional models by focusing on generating data first, and only then using the generated data for information extraction. Therefore, the data generation process based on the ACDF model consists of two main stages:

Stage 1: Data Preprocessing. In this stage, the paper uses a technique to aggregate and group contextual information by filtering out important information and self-adjusting to fit data with varying numbers of attacks, thereby effectively maintaining context and detecting abnormal patterns.

Stage 2: Synthetic Data Generation. Based on the results from the data preprocessing stage, a Diffusion model applying UNet1DModel with set parameters, including the number of diffusion steps, is used to generate new attack data samples that replicate hacker tactics. This process enhances the richness and diversity of the data, balancing the number of minority attack samples by combining real data with generated data, thus minimizing bias and improving the effectiveness of detecting APT threats.

Second, regarding the feature extraction and aggregation problem This study proposes using the mLSTM model to analyze and detect APT attacks. With mLSTM, the model leverages deep learning capabilities through an attention mechanism, integrates Conv1D to process local information, and uses Learnable Skip Connections and Group Normalization to optimize data flow. As a result, mLSTM efficiently exploits sequential relationships and complex structures among data flows while focusing on important features, thereby enhancing the model's detection accuracy.

1.3 Contributions of the paper

- Successfully developed a novel Ensemble Learning model based on the combination of data generation and feature extraction techniques. The proposed ACDF-mLSTM model is based on new ideas and methodologies, departing from conventional architectures and approaches. The results in the experimental section demonstrate that the ACDF-mLSTM model performs well not only on a single imbalanced dataset but also across various datasets under multiple evaluation scenarios and parameter configurations. This not only proves the scientific significance of the paper but also showcases its practical effectiveness.
- Proposed a novel data generation method based on the ACDF model. With the data processing and generation approach within the ACDF model, this paper demonstrates for the first time that selecting and generating data before aggregation and extraction can yield effective results, but it requires careful attention to aggregating contextual information to minimize noise and eliminate redundant, unimportant data.
- Proposed the use of mLSTM for the task of APT IP feature aggregation and extraction. Evidently, based on this proposed method, the ACDF-mLSTM model significantly

reduced computational and processing time compared to models based on other aggregation platforms, while still maintaining high accuracy. This result opens a new direction for applying the mLSTM model to other network attack detection tasks.

2 Related works

The study by Fuguang Bao et al. [17] addresses the significant challenge of anomaly detection in high-dimensional and imbalanced data within wireless sensor networks (WSNs). This data is often large-scale, computationally complex, unevenly distributed (with anomalous data being very rare), and possesses sequential relationships. To overcome the limitations of traditional methods, the paper proposes a new anomaly detection network model: kNN-SMOTE-LSTM. This model is designed based on a combination of three main components: an LSTM network serves as the base classifier due to its ability to efficiently handle time-series data and learn long-term dependencies; experiments showed LSTM achieved the highest AUC value (0.9132) compared to other classifiers. The SMOTE algorithm is applied to augment the minority class data, addressing the overfitting problem caused by severe data imbalance. To mitigate the issue of SMOTE potentially generating noisy data that affects the classification boundary, a kNN discriminant classifier is intelligently integrated with LSTM to filter out valid samples and remove noisy ones, ensuring only "safe" samples are added to the training set, thereby significantly improving classification performance. Experimental results demonstrate that the kNN-SMOTE-LSTM model significantly improves the classification performance on imbalanced sequences, achieving an F-score of 0.9167 and an AUC of 0.9296, outperforming other oversampling methods when combined with LSTM. The model not only enhances the accuracy for minority samples but also addresses the misclassification of majority samples, with a misclassification rate for most samples of only 0.000082 and an overall accuracy of 99.97%.

The research by K. Saranya and A. Valarmathi [22] proposes the Multi-Layered Deep Autoencoder (M-LDAE), a novel network model designed to address the critical challenge of detecting complex cross-layer attacks in IoT networks. To overcome the limitations of traditional techniques in feature representation, scalability, and flexibility, M-LDAE extracts latent representations containing both global and local attributes through the hierarchical simplification capability of deep autoencoders. The model integrates advanced deep learning algorithms such as Recurrent Neural Networks (RNNs), Graph Neural Networks (GNNs), and Temporal Convolutional Networks (TCNs) to enhance the detection of time-series, structured, and sequential threats, as well as to adapt to emerging attack methods. The system uses a reconstruction error-based anomaly detection technique, combined with ensemble learning and thresholding to reduce the false positive rate and increase detection sensitivity. M-LDAE operates by analyzing data at multiple protocol layers, enabling the detection of attacks like Man-in-the-Middle at the network layer and Distributed Denial of Service (DDoS) at the transport layer. Extensive testing on standard datasets (NF-ToN-IoT, NF-UNSW-NB15) and realistic scenarios has demonstrated that M-LDAE can adapt to new attack vectors, enhance detection accuracy, and minimize false positives, achieving an accuracy of 94.8% and a false positive rate of 6.8%, while outperforming existing methods like RNN, GNN, and TCN across all key performance indicators. This confirms M-LDAE as a robust and flexible security solution for various IoT domains.

The study in [23] addresses the severe data imbalance problem in attack detection for IDS using Machine Learning and Deep Learning. Attack data often has an uneven distribution, causing IDS models to be biased and inefficient at detecting minority attack classes. To overcome the limitations of traditional methods like sampling or SMOTE, the paper proposes a new model combining a Wasserstein Conditional Generative Adversarial Network (WCGAN) and an XGBoost Classifier. The model uses a Deep Auto-Encoder (DAE) to extract important features, after which WCGAN generates synthetic attack data for minority classes using a gradient penalty for stable learning. Finally, the XGBoost Classifier is used for intrusion detection. Evaluation on the NSL-KDD, UNSW-NB15, and BoT-IoT datasets shows that WCGAN is more stable than other GAN variants. More importantly, when using mixed data (real and synthetic), the proposed model significantly improved accuracy, recall, and F1-score, outperforming similar techniques and demonstrating its ability to effectively handle the data imbalance problem for minority attack classes.

The research in [24] is a systematic review of the role of Generative Adversarial Networks (GANs) in network anomaly detection, focusing on data representation learning rather than just data augmentation. The core problems in anomaly detection are the scarcity of anomalous data, reliance on known attack knowledge (making it difficult to handle zero-day attacks), and data imbalance. GANs, with their ability to generate new, realistic data and learn data distributions, have emerged as a promising solution. Advanced GAN models were reviewed for how they address challenges such as reducing dependency on prior knowledge, improving resource efficiency, and handling data imbalance. Despite significant academic progress, the practical implementation of GANs in live network environments remains very limited due to difficulties in collecting diverse data, adapting to changing network architectures, and privacy concerns.

Future research directions include developing real-time training and detection, enhancing resource efficiency, optimizing hyperparameters, integrating zero-shot learning for detecting unseen anomalies, and improving transfer learning capabilities.

The study in [25] proposes a new intrusion detection algorithm to address the severe data imbalance in IDS, which causes high false detection rates and poor performance for minority attack types. The proposed model combines an enhanced random forest with the SMOTE algorithm. Specifically, in the preprocessing stage, data is balanced using a hybrid sampling method that combines K-means clustering and SMOTE, which increases the number of minority samples, enriches feature attributes, and minimizes outliers. The processed data is then fed into the enhanced random forest model, where decision trees with superior classification performance are selected, and trees with high similarity or poor performance are removed to build a more robust forest. Finally, the preliminary prediction results are further calibrated using an attack-type similarity matrix and predefined operational rules to ensure more rational decisions. When evaluated on the NSL-KDD dataset, the model achieved an accuracy of 99.72% on the training set and 78.47% on the test set. More importantly, this method showed significant improvement in accuracy, recall, and F1-score, especially in detecting minority attack classes like U2R and R2L, proving its effectiveness in solving the data imbalance problem. Future work will focus on optimizing accuracy and computation time through feature selection and classifier choice.

The research in [26] presents an IDS for Open Radio Access Networks (O-RAN), focusing on solving the data imbalance problem where minority attack types are often overlooked. To address this, the authors used sampling techniques like SMOTE and variants of Generative Adversarial Networks (GAN) to create synthetic data. The system was evaluated on two datasets: CICEVSE2024 and a real-world Open RAN dataset. The results showed that SMOTE variants improved the accuracy of weaker classifiers, especially Naive Bayes (NB), on the CICEVSE2024 dataset. However, synthetic data from GANs (CTGAN) did not significantly improve the detection performance of traditional machine learning classifiers. Notably, applying SMOTE sometimes even reduced the accuracy of some classifiers on the RAN dataset, possibly due to overfitting or the introduction of noisy data. The study acknowledges limitations such as its focus on binary classification and the challenge of ensuring synthetic data accurately reflects real network traffic. Future research directions include expanding to multi-class classification and exploring more advanced hybrid and deep learning models.

The study in [20] proposes a new strategy for IDS based on a Conditional Generative Adversarial Network (cGAN) to address the severe data imbalance in network traffic, where minority attacks are rare and often missed. cGAN is used to generate synthetic attack samples with a distribution close to the real data, expanding the minority classes without causing information redundancy. The strategy includes four main parts: a generator module (G), a discriminator module (D), a target network (F), and an identification network (FClassification). Experimental results on the NSL-KDD dataset showed an overall accuracy of 98.65% in 23-class classification, significantly improving metrics like precision, recall, and F-value. On the UNSW-NB15 dataset, this method also showed high generalization ability, achieving the highest detection rates for the Backdoor (0.82) and Worms (0.92) classes. In summary, this method effectively solves the data imbalance problem, significantly enhancing the detection efficiency and generalization ability of the IDS model.

The research in [18] introduces the MCG hybrid learning model for APT attack detection, designed to overcome the limitations of traditional methods such as poor generalization and difficulty handling imbalanced data. MCG integrates a Multilayer Perceptron (MLP) for flow feature processing, Correlated Recursion (CR) to build an IP information network, and a Graph Attention Network (GAT) to analyze IP relationships, while effectively addressing data imbalance by highlighting minority attack classes. Experiments show that MCG achieves outstanding performance, with an accuracy of up to 99% and precision, recall, and F1-score values reaching near-perfect levels (1.00, 0.99, and 0.99, respectively). This model has proven to be significantly more effective at detecting APTs on imbalanced data compared to existing methods, offering a reliable solution for early threat detection.

The study in [6] proposes BGCRKAN, an advanced hybrid deep learning model aimed at enhancing APT attack detection by addressing data imbalance and network complexity. The model combines an improved Kolmogorov-Arnold Network (KAN) using Catmull-Rom splines to dissect complex APT attack behavior and handle uneven data distributions. It also integrates a bidirectional graph attention mechanism (BGATv2) to capture multi-step and concealed attack features by analyzing forward and backward information flows between nodes. Using data constructed from public threat intelligence sources, BGCRKAN demonstrated superior performance, achieving an accuracy of 97.10% with a low False Positive Rate (FPR) of 0.2% and a False Negative Rate (FNR) of 9.02%. Although the model has a tendency to generate false positives and has higher computational costs for very large graphs, BGCRKAN provides a robust solution for early APT detection.

The research in [12] proposes a new hybrid deep learning method for detecting network attacks in IoT environments. The model combines Self-Organizing Maps (SOMs), Deep Belief Networks (DBNs), and Autoencoders, optimized by Particle Swarm Optimization (PSO) and an advanced optimization algorithm. The goal is to identify both known and unseen threats. The method also integrates an attention mechanism to enhance feature extraction and a new cost function for performance evaluation. Experiments on the NSL-KDD, UNSW-NB15, and CICIoT2023 datasets showed strong performance, achieving accuracy up to 99.99% and a Matthews Correlation Coefficient (MCC) exceeding 99.50%. This demonstrates a significant potential to improve IoT security by adapting to evolving attack strategies.

Finally, the study in [27] proposes a GCN-based APT detection method to address the challenge of complexity and stealthiness in these attacks. The model is built by extracting organization-vulnerability relationships from APT threat intelligence and software security entities from CVE, CWE, and CAPEC to create a knowledge graph of APT attack behavior. This knowledge graph is then converted into a homogeneous graph, and a GCN is used to process the graph features to effectively detect APT attacks. The method was evaluated on a self-constructed dataset, showing that the GCN's detection accuracy reached 95.9%, an improvement of about 2.1% over the GraphSAGE method. This demonstrates the approach's effectiveness in realistic APT detection scenarios, especially with larger datasets, and significantly enhances information system security.

3 The ACDF-mLSTM Model for APT detection

3.1 Proposed model architecture

Figure 1 illustrates the architecture of the ACDF-mLSTM model. Its main components include:

- Adaptive Contextual Information Aggregation and Collection: This block utilizes AC as an "intelligent filter" to select the most critical information from chaotic network data. It also flexibly self-adjusts to perform well regardless of the number of attacks in the data, helping to preserve the surrounding context for the model to better understand the relationships within the data.
- Generating Synthetic Data to Simulate Attack Data in Networks: In this block, the Diffusion Model (DF) relies on the information filtered by AC to create new attack data samples, making the dataset more diverse and less skewed. This provides the model with more varied data to learn from and detect attack tactics.
- Feature Extraction and Aggregation: This component uses mLSTM, which stores data in a matrix format and employs a query, key, and value mechanism to gain a holistic view, rather than focusing on individual small samples. This enables it to detect signs of attack based on the bigger picture. With the query, key, and value mechanism, mLSTM can ignore noise and focus only on the most valuable parts of the data. It can process long and complex data sequences without information loss, ensuring accurate analysis.
- Classification: This block classifies IP-Pairs based on the data that has been processed and balanced.

3.2 Generating synthetic data simulating APT attacks in networks based on ACDF

3.2.1 Theoretical basis of AC

AC is a method for time-series data analysis, inspired by the research of Fawaz et al. [28]. This method focuses on identifying important points within sequential data and analyzing context by considering preceding and succeeding data points. This allows the deep learning model to recognize complex patterns and contextual relationships, thereby improving its classification ability. The process begins by identifying important data points to serve as centers, then creating a contextual window by taking the data points before and after the center point. The relationships and interactions between these points are analyzed to identify abnormal patterns. Information from the points within the contextual window is combined using deep learning techniques, creating a representative frame for the entire context. This frame is used to detect hidden patterns and relationships, helping to clearly distinguish between normal and abnormal behaviors. The principle for creating these contextual windows is expressed in formula (1):

$$y_t = \{Central(x_t) + Contextual(x_{t-k}, \ldots, x_{t+k}) + b\} \tag{1}$$

where:

- y_t: The data frame at time t.
- $Central(x_t)$: The important data point set as the center at time t
- $Contextual(x_{t-k}, \ldots, x_{t+k})$: The context surrounding the data center at time t
- b: A bias value to adjust the result.

3.2.2 Theoretical basis of the diffusion model

The Diffusion Model in this research, used for generating synthetic data simulating network attacks, is built on the principle of gradual diffusion. This is an advanced data generation method designed to create high-quality data samples with fine-grained control over the generation process. The structure of the Diffusion Model consists of the following main components:

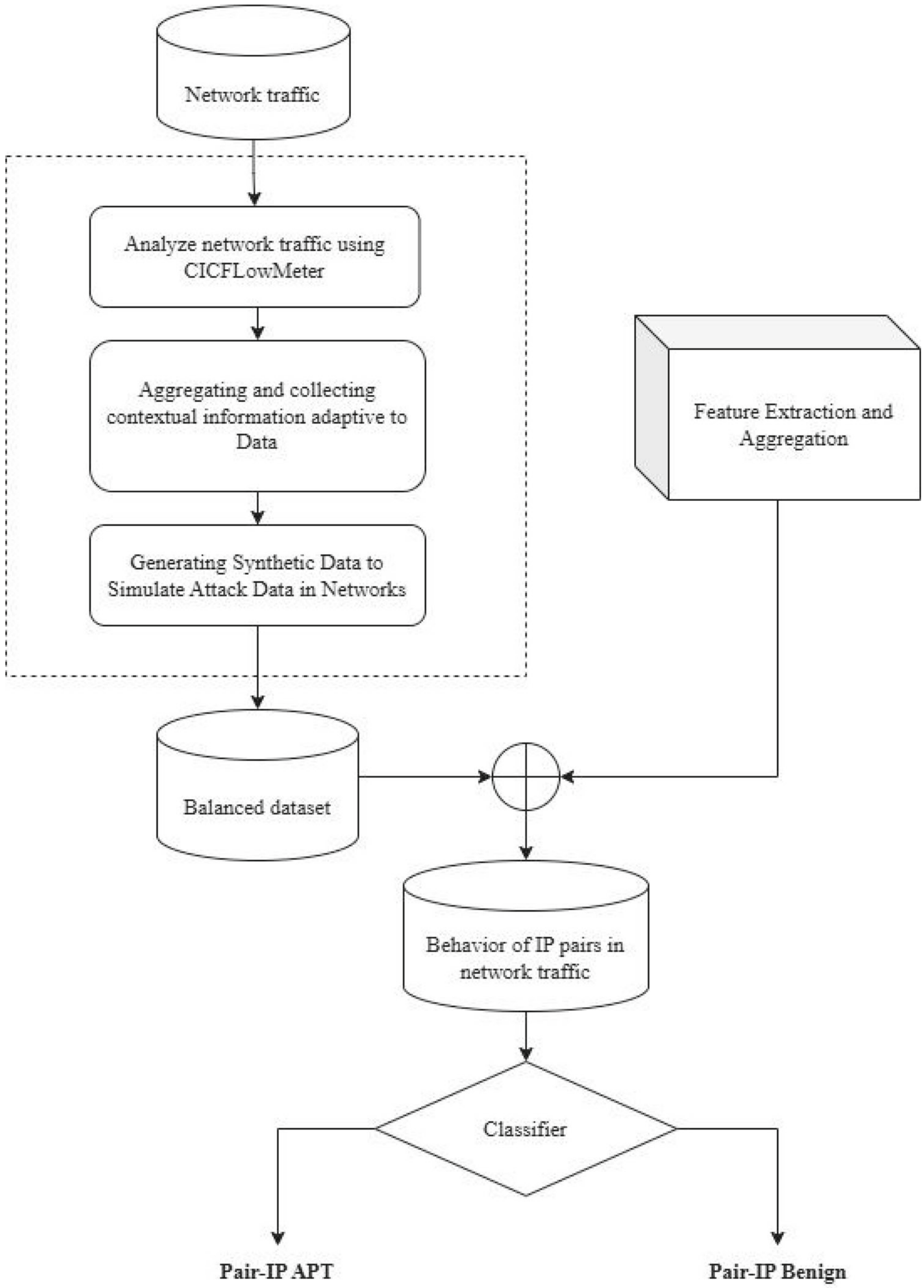


Fig. 1 Architecture of the ACDF-mLSTM model for APT attack detection

- Generator (Reverse Diffusion Process): The input to this process is a random noise vector, typically sampled from a Gaussian distribution after undergoing many forward diffusion steps. Along with it is a condition vector *c* containing additional information such as labels or context. The task of the reverse diffusion process is to restore the original data from the noise, using a neural network to gradually remove the noise and transform it into new data ($\times$ 0 from xt), with the goal of creating synthetic data that closely resembles the real data based on the condition.
- Discriminator (Forward Diffusion Process): The forward diffusion process takes real data *x* and gradually adds noise according to a defined schedule (usually Gaussian noise), along with the condition information *c*. Its role is to simulate how data is transformed into complete noise over many steps. The neural network in this process learns to predict the noise level at each step, helping the reverse process understand how to restore the data.
- The loss function is described by formula (2):

$$L = E\left[\left\|x_o - \hat{x}_o\right\|^2\right] \quad (2)$$

Where:

- $E[\|x_0 — \hat{x}_0\|^2]$: The expectation of the squared error between the original data x_0 and the reconstructed data $\hat{x}_0$ after the reverse diffusion process.
- The training process is based on optimizing this error over many diffusion steps.

The training of the Diffusion Model is akin to a gradual denoising process. The objective is to train the model to learn how to reverse the forward diffusion process, restoring meaningful data from random noise. This helps the model generate high-quality synthetic attack samples, ensuring diversity and relevance to real-world contexts.

3.2.3 Adaptive Contextual Information Aggregation and Collection based on AC

The AC model performs adaptive contextual information aggregation and collection, as illustrated in the figure presented in the paper [29], consisting of the following steps:

- Step 1: Identify Focused Data and Context: The process begins by identifying important data points (Central) and their surrounding context (Contextual). Each flow is analyzed to find significant flow points, which then form centers. A sufficiently large data frame is then created around each center, including the data points before and after it, to form a frame containing all important data and its surrounding context.
- Step 2: Data Analysis: After identifying the important data centers and their surrounding context, the analysis process begins. The goal of this step is to analyze data points that are far from the center and outside the contextual region. This ensures clear identification of which data is likely noise to be removed, and which data, despite being distant, still holds value and should be added to the frames for analysis in subsequent steps.
- Step 3: Eliminate Redundant Data: Once the parts to be removed have been analyzed and identified, this step proceeds to discard noisy data, which is essentially unnecessary contextual information. This process helps optimize resources and focus on the information relevant to the problem, while also reducing data complexity.
- Step 4: Data Aggregation: Finally, the remaining data parts are aggregated into a complete data frame. This data is smaller in size than the original but still ensures it contains all detailed information of the components. It even removes noise-inducing elements, helping the model learn the focused data and its surrounding context. This allows for the analysis of surrounding impact relationships and mutual dependencies of the data, creating a more comprehensive view for the model later on.

Based on these four steps, we find that this approach offers many benefits to the system. The process of identifying and focusing on important data points along with their context gives the model a deeper and more comprehensive view of the data. Eliminating redundant and noisy data not only optimizes resources but also improves the model's accuracy and performance. By aggregating the data into a complete frame, the model can easily identify and analyze complex relationships and interdependencies among data components. This creates a solid foundation for detecting and predicting abnormal patterns or potential threats, thereby enhancing the system's reliability and effectiveness in handling complex situations.

3.2.4 Data generation based on the diffusion model

The operational principle of the Diffusion Model used for generating synthetic data is described in Figure 2 below.

The Diffusion model generates synthetic data simulating network attacks, as described in Figure 3, consisting of the following steps:

- Step 1: Prepare Conditional and Noise Data: The process begins by creating a random noise vector from a Gaussian distribution, representing the initial state of the data in the latent space. Simultaneously, a condition vector is prepared, containing additional information such as classification labels. This condition vector is processed through preprocessing steps, such as a Flatten layer to flatten the data and an Embedding layer to convert it into a suitable vector representation. Subsequently, the noise vector and the condition vector are integrated through mechanisms like attention or concatenation, forming the input for the Diffusion model. This step ensures that the conditional information will guide the subsequent denoising process.
- Step 2: Generate Synthetic Data: Using a U-Net architecture with attention blocks, the model performs a gradual denoising process starting from the initial noise vector. This process occurs over many iterative steps, where the model removes noise based on the conditional information and previously trained parameters. The result is synthetic data with corresponding labels, reflecting the actual distribution of attack and normal behaviors in the network. To enhance accuracy, the number of denoising steps can be adjusted, or model parameters can be fine-tuned to improve the authenticity of the generated data.
- Step 3: Model Optimization: The Diffusion model is trained to minimize a loss function, typically the mean squared error (MSE) between the predicted noise and

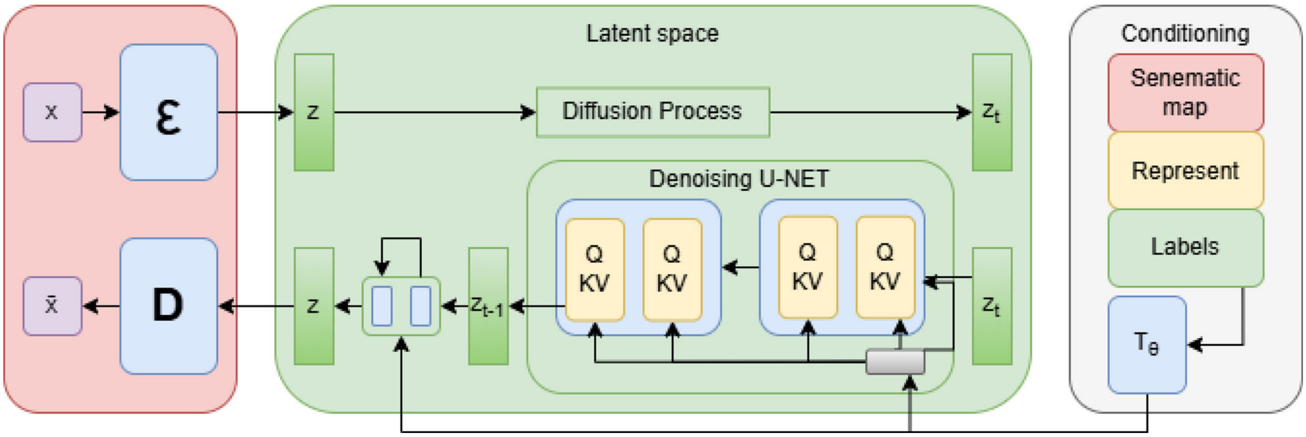
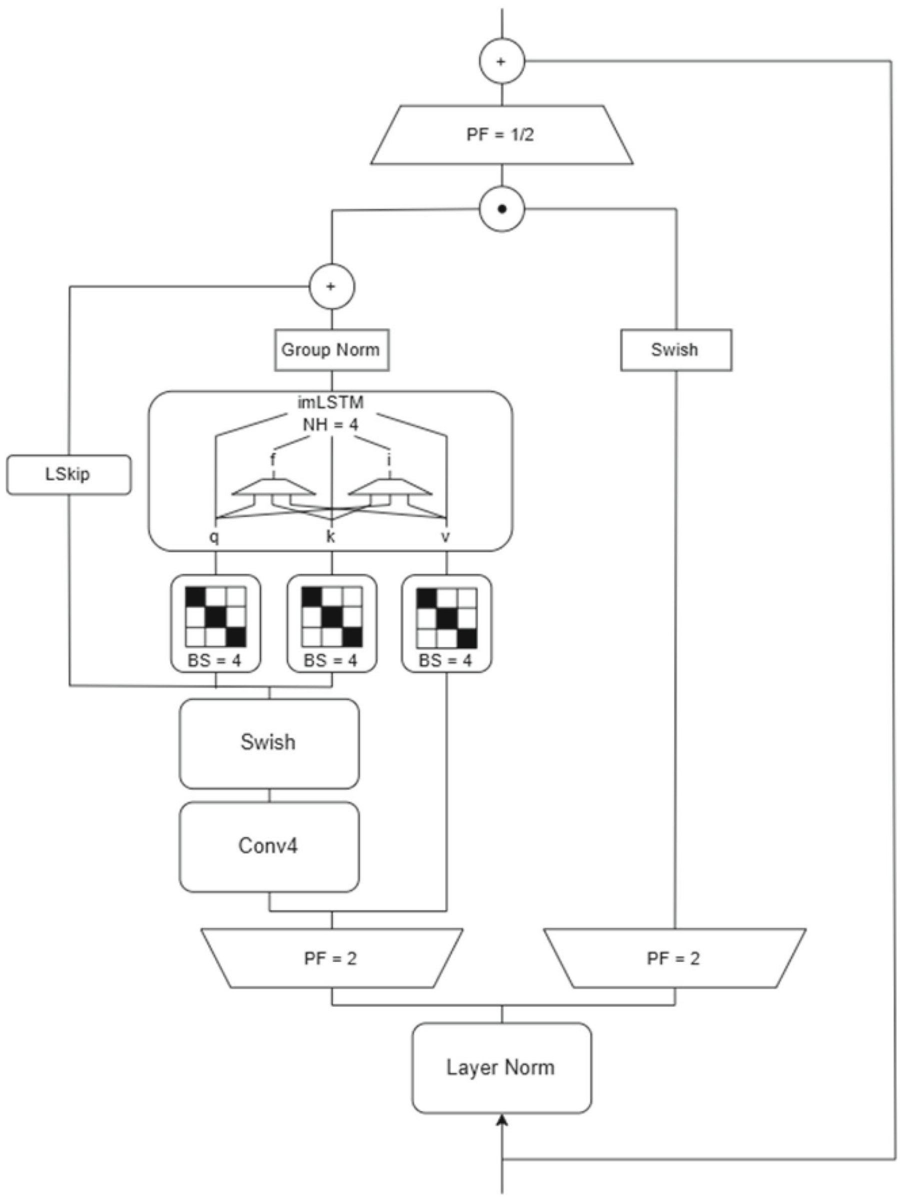


Fig. 2 The process of synthetic data generation by the diffusion model

Fig. 3 Operational principle of mLSTM



the actual noise added at each denoising step. This process does not use a discriminator or adversarial training; instead, the model learns directly from the data by improving its ability to predict and remove noise through training loops. Training continues until the model converges, ensuring that the generated synthetic data has characteristics close to the real data.

- Step 4: Create a Balanced Dataset: After the Diffusion model is optimized, the synthetic data is combined with the real data to create a balanced dataset. Techniques such as resampling or weighted sampling can be applied to ensure that minority classes (like attack data) are fully represented. This balanced dataset enhances diversity and improves the generalization ability of machine learning models in detecting complex attack patterns. Cross-validation can be performed to evaluate the performance of the new dataset and confirm its effectiveness. This procedure fully leverages the Diffusion model's capability to generate high-quality synthetic data, effectively supporting the simulation of complex network attack patterns and enhancing the system's detection capabilities in real-world environments.

3.3 Feature Extraction and Aggregation based on mLSTM

3.3.1 Theoretical Basis

The mLSTM model [30] surpasses the traditional LSTM model in its parallel storage and computation capabilities. Unlike LSTM, mLSTM shifts from storing data as a vector to storing it in a matrix format. For information retrieval, the model also uses query, key, and value, similar to the Attention mechanism. Formula (3) calculates these values as follows:

$$\begin{aligned}
\mathrm{q_t} &= \mathrm{w_q x_t} + \mathrm{b_q}\\
\mathrm{k_t} &= \frac{1}{\sqrt{\mathrm{d}}}\mathrm{w_k x_t} + \mathrm{b_k}\\
\mathrm{v_t} &= \mathrm{w_v x_t} + \mathrm{b_v}
\end{aligned} \tag{3}$$

where: $W_{q,k,v}$: are the weight matrices for query, key, and value, respectively $b_{q,k,v}$: are the bias coefficients for query, key, and value, respectively

This also changes the update formulas for the cell state, normalizer state, and hidden state, as shown in formula (4).

$$\begin{aligned}
C_t &= f_t C_{t-1} + i_t v_t k_t^T\\
n_t &= f_t n_{t-1} + i_t k_t^T\\
h_t &= O_t \Theta^{h}_{-t}\\
\underline{h}t &= c_t q_t \Big/ \max\left\{\left|n_t^T q_t,\ 1\right|\right\}
\end{aligned} \tag{4}$$

Using a storage capability similar to the Attention mechanism allows the model to learn the relationship between value and key through the new cell state update formula. Simultaneously, the hidden state can also query necessary data by using the query.

3.3.2 Feature extraction and aggregation based on mLSTM

The operational principle of the combined mLSTM model for feature extraction and aggregation is described in Fig. 3:

The mLSTM model performs feature extraction and aggregation as described in Figure 3 consisting of the following steps:

- Step 1: The input data is normalized and then passed through an Up-Projection Factor layer with a dimension expansion factor of 2 to expand its dimensionality.
- Step 2: After dimension expansion, the data is split into two parallel processing branches:

The first branch feeds the data into the mLSTM model. The data is parallelized into two streams:

- One stream passes through a Conv1D layer with 4 kernels for information aggregation, then through a Learnable Skip Connection layer and a matrix diagonalization layer with 4 heads for the query and key.
- The remaining stream is diagonalized with 4 heads, serving as the value in the attention mechanism.

The second branch processes the data through a simple MLP with a Swish activation function, acting as an external output gate to control the information flow from the mLSTM block.

- Step 3: The query, key, and value components are fed into the mLSTM model to compute attention. The model's output is then normalized using Group Normalization. This normalized result is added to the output from the Learnable Skip Connection layer.
- Step 4: The output, after being added to the skip connection, is dot-multiplied with the result from the external output gate and then passed through a Down Projection Factor layer to restore the data's original dimensionality. Finally, the output is added to the original input data and passed on to the next block in the model.

The mLSTM process efficiently leverages feature extraction and aggregation capabilities through parallel processing and an attention mechanism, ensuring flexible and accurate information control. The integration of Learnable Skip Connection and Group Normalization helps optimize the output quality, providing strong support for applications requiring complex data processing in real-world scenarios.

3.4 APT IP classification

For the classification block, we use Softmax Layers. The Softmax layer is responsible for calculating the probability for each class, APT or Benign, which helps determine the likelihood of an input belonging to a specific class. Softmax computes these probabilities by transforming the output values into probabilities within the range of [0, 1], ensuring that the sum of all probabilities equals 1. Formula (5) expresses how the probability for class i is calculated:

$$P_i = \frac{expexp(z_i)}{\sum_{j=1}^{c} exp\left(z_j\right)} \tag{5}$$

where:

- P_i is the probability that the input belongs to class i.
- z_i is the output from the Fully Connected layer corresponding to class i.
- C is the total number of classes.

4 Experiments and evaluation

4.1 Experimental data

The dataset consists of both malicious and clean network traffic. The malicious portion was collected and analyzed from 29 files in the Malware Capture-13 dataset, comprising 6 types of malware used in APT attacks: Andromeda, Colbalt, Cridex, Dridex, Emotet, and Gh0stRAT. The clean network traffic was extracted from the e-government server of Quang Nam province as part of the scientific research project KC.01.05/16–20, funded by the Vietnam Ministry of Science and Technology. This dataset was collected over a single day on July 27, 2019.

In addition to evaluating on our own IP classification dataset, we will also perform evaluations on three public datasets using a binary classification approach: NF-BoT-IoT-v2, NF-ToN-IoT-v2, and NF-CSE-CIC-IDS2018-v2. The information for these three datasets is as follows:

- NF-BoT-IoT-v2: An extension of the NF-BoT-IoT dataset, updated from 8 to 43 features. The dataset was constructed based on network traffic monitoring protocols in a 2018 network environment and includes 37,763,497 flows, with 37,628,460 being malicious and 135,037 being benign.
- NF-ToN-IoT-v2: An extension of the NF-ToN-IoT dataset, updated from 8 to 43 features. The dataset was built using a network traffic monitoring protocol and contains a total of 16,940,496 flows, with 10,841,027 being malicious and 6,099,469 being benign.
- NF-CSE-CIC-IDS2018-v2: An extension of the NF-CSE-CIC-IDS2018 dataset, updated from 8 to 43 features. The dataset is based on network traffic monitoring protocols and was collected by the University of Queensland in 2018. It contains a total of 18,893,708 flows, with 2,258,141 being malicious and 16,635,567 being benign.

Details of the datasets are listed in Table 1 below:

4.2 Experimental scenarios and evaluation

4.2.1 Experimental method

To ensure that the performance differences are not dependent on a particular data split or random initialization, we adopted the following methodology. First, the data was split into a training set (80%) and a test set (20%). This process was repeated 10 times, each with a different random partition, to mitigate the influence of randomness in data splitting or model parameter initialization. The results reported in the paper are the average values from these 10 runs. This approach provides a more reliable estimate of the model's true performance while minimizing the variance caused by stochastic factors (Table 2).

4.2.2 Experimental scenarios

Several experimental scenarios will be conducted in this paper to demonstrate the model's effectiveness and to answer the following research questions (RQs):

- RQ1: How effective is the ACDF-mLSTM model in the task of APT IP detection? In this scenario, the paper will fine-tune the parameters of the ACDF-mLSTM model to determine which parameter settings yield the best results.
- RQ2: How does the ACDF-mLSTM model address the data imbalance problem compared to other research and approaches? To answer this question, the paper will compare and evaluate the ACDF-mLSTM model against several other studies and approaches that use data balancing techniques. Specifically, some baseline models for comparison will include: BiLSTM-Attention-Dropout [18], KNN-SMOTE-LSTM [17], and VAE-LSTM [19].
- RQ3: How has the ACDF-mLSTM model improved the ability to aggregate and classify APT IPs compared to other research and approaches? To achieve this objective, we selected several baseline models based on natural language processing and graph-based techniques for evaluation, including: BiLSTM-Attention [19], GCN-BiLSTM [31], and MIG [32].

Table 1 Statistics of the experimental datasets

Full	Type	Total	APT	BENIGN
Malware Capture	Flows	1,671,393	871,914	799,479
	Pair-IP	1,884	218	1,666
NF-BoT-IoT-v2	**Flows**	37,763,497	37,628,460	135,037
NF-ToN-IoT-v2	**Flows**	16,940,496	10,841,027	6,099,469
NF-CSE-CIC-IDS2018-v2	**Flows**	18,893,708	2,258,141	16,635,567

Bold indicates data differences

Table 2 Selecting and configuring the hyper-parameters of the ACDF-mLSTM framework

Model	Hyper-parameter	Value
AC	Window	10
Diffusion Model	Sample size	64
	In channels	128
	Out channels	128
	Layers per block	1
	Block out channels	(32.64)
	Mid block type	UNetMidBlock1D
	Down block types	DownBlock1D
	Up block types	UpBlock1D
	Number of diffusion steps	1000
	Learning rate	1e-4
mLSTM	Input embedding size	128
	Hidden size	256
	Number of layers	2
	Number of attention heads	4
	Conv1D filters	4
	Activation function	Swish Learning
	Learning rate	1e-4
Dropout	Dropout probability	0.1
Training	Batch size	32
	Temperature	0.1
	Learning rate	1e-4
	Loss function	MSE
Classifier	Activation	Softmax
	Loss function	Cross-entropy

4.3 Evaluation criteria and parameter configuration

4.3.1 Evaluation criteria

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{6}$$

where :

- TP—True positive: The number of malicious samples classified correctly.
- FN—False negative: The number of malicious samples classified as normal.
- TN—True negative: The number of normal samples classified correctly.
- FP—False positive: The number of normal samples classified as malicious.

Precision:

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{7}$$

Recall:

$$Recall = \frac{TP}{TP + FN} \times 100\% \tag{8}$$

F1-score:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{9}$$

4.3.2 Parameter configuration

4.3.3 Installation requirements

Some information about hardware and software requirements to be able to train and test this model is as follows:

Hardware

- CPU: Multi-core processor (e.g. Intel Core i7 or equivalent) to support parallel computing.
- RAM: Minimum 16GB to handle large datasets and model training processes.
- GPU: Graphics card with at least 4GB VRAM (e.g. NVIDIA GTX 1060 or equivalent) to accelerate deep learning model training.
- Storage: Minimum 500GB storage capacity to hold PCAP data and trained models.

Software

- Operating system: Linux (e.g. Ubuntu 20.04) or Windows 10.
- Programming language: Python 3.8 or higher.
- Libraries and frameworks:
  - TensorFlow 2.x or PyTorch 1.x to build and train the model.
  - Scikit-learn for data preprocessing and model evaluation.
  - CICFlowMeter to extract features from PCAP files.
  - Pandas and NumPy for data processing.
- Version management tools: Git to track source code and models.

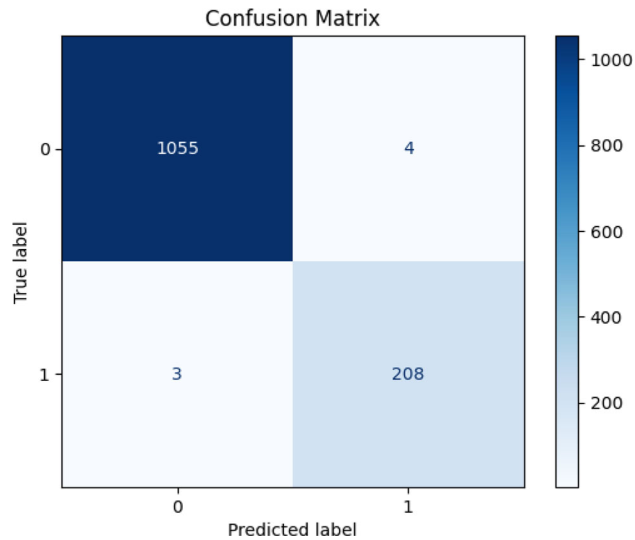


Fig. 4 Confusion matrix of the ACDF-mLSTM model with the best result

4.4 Experimental results

4.4.1 Experimental results for scenario 1

Table 3 below presents the experimental results when varying the parameters of each component in the ACDF-mLSTM model. It should be noted that the results shown in this table are the best outcomes selected from the experimental process.

From the results in Table 3, it is evident that the ACDF-mLSTM model demonstrates outstanding performance, achieving an accuracy of up to 99.80, precision of 98.50, recall of 99.00, and an F1-score of 98.75% in its optimal configuration. These results indicate the model's ability to detect APT attack patterns accurately and effectively, especially when comparing different model configurations. The reasons for this impressive performance can be explained by the following factors:

First, the Diffusion model, with its flexible block_out sizes (32, 64, 128), plays a crucial role in extracting high-quality features. In the optimal configuration (block_out = 64), the Diffusion model uses UNet1DModel with two down/up-sampling blocks and a maximum of 64 channels, allowing the model to effectively remove noise and focus on important data features. This helps create a rich latent space, supporting the detection of sophisticated attack patterns. Specifically, the configuration with block_out = 64 and Window = 10 achieved an F1-score of 98.75%, which is higher than block_out = 32 (F1-score 97.80%) or block_out = 128 (F1-score 98.00%), indicating that block_out = 64 is the optimal balance between complexity and feature aggregation capability.

Second, mLSTM, with its attention mechanism (4 heads), Conv1D, Learnable Skip Connection, and Group Normalization, enhances the ability to process long-term context and complex relationships between flows. In the configuration of Window = 10 and mLSTM Layers = 2, the model achieved its highest performance (accuracy 99.80%) due to its ability to parallelize data through two branches: one using Conv1D to aggregate local information and attention to focus on important features, and the other using an MLP with a Swish function as an output gate to control information flow. This combination helps the model capture the overall data landscape, especially with Window = 10 (considering 10 flows before and after the APT center), allowing for deeper analysis compared to Window = 5 (accuracy only 99.05–99.10%). The use of Learnable Skip Connection and Group Normalization also ensures stable output, preventing the loss of important information.

Comparing the configurations, the setup with Window = 10, Diffusion block_out = 64, and mLSTM Layers = 2 outperforms the others. With Window = 5 and block_out = 32, the performance is lower (accuracy 99.05, F1-score 97.80%) due to the limited number of flows and small block_out size, which prevents the model from fully exploiting context and features. Similarly, the configuration with block_out = 128 and mLSTM Layers = 3 (accuracy 99.20, F1-score 98.00%), while an improvement over Window = 5, does not achieve the high performance of block_out = 64, possibly due to increased complexity leading to overfitting or an imbalance in feature processing. The balance between Window = 10 and block_out = 64 allows the model to leverage a wider context while maintaining computational efficiency.

To better understand the model's performance, let's consider the Confusion Matrix. Figure 4 below provides a detailed view of these values.

The results in Fig. 4 show that the ACDF-mLSTM model exhibits impressive classification performance, with 1055 samples correctly predicted as label 0 (True Negative),

Table 3 APT attack detection results of the model with different parameters

Model			Evaluation of IP			
ACDF-mLSTM			Acc	Pre	Rec	F1
Window (Flows before/after APT center)	Diffusion block_out	mLSTM Layers				
5	32	1	0.9905	0.9755	0.9805	0.9780
5	64	2	0.9910	0.9770	0.9820	0.9795
10	**64**	**2**	**0.9980**	**0.9850**	**0.9900**	**0.9875**
10	128	3	0.9920	0.9785	0.9815	0.9800

Bold indicates the superior performance of the proposed method

demonstrating an excellent ability to identify non-attack samples. At the same time, the model also correctly predicted 208 samples as label 1 (True Positive), proving its effectiveness in detecting APT attack samples despite their significantly smaller number. Notably, there were only 4 samples of label 0 mistaken for label 1 (False Positive) and 3 samples of label 1 mistaken for label 0 (False Negative), which testifies to the model's stability and high accuracy, thanks to the Diffusion mechanism with an optimal block_out size (like 64) that helps remove noise and extract high-quality features.

The balance between the two classes in this matrix further highlights the strength of mLSTM, with its attention mechanism (4 heads) and Learnable Skip Connection, allowing the model to accurately capture long-term context and complex relationships between data flows, especially when applying Window = 10. However, with the number of True Positives (208) being much lower than True Negatives (1055), the recall for label 1 may need further improvement, perhaps by further fine-tuning parameters such as the number of mLSTM layers or diffusion steps. Nevertheless, the extremely low error rate (only 7 misclassified samples out of more than 1200 total) is clear evidence of the superiority of ACDF-mLSTM.

4.4.2 Experimental results for scenario 2

The results presented in Table 4 below are from the comparison and evaluation of the proposed model against several other studies, aiming to answer the research question: How does ACDF-mLSTM demonstrate its data synthesis and generation capabilities compared to other research?

Based on Table 4, the ACDF-mLSTM model achieves outstanding performance in detecting APT attacks when handling imbalanced data, thanks to the use of the Diffusion model for synthetic data generation. With an accuracy of 99.80, precision of 98.50, recall of 99.00, and an F1-score of 98.75% in its optimal configuration, ACDF-mLSTM demonstrates clear superiority over VAE-LSTM and KNN-SMOTE-LSTM. This is attributed to the following factors:

First, diverse synthetic data generation: The Diffusion model uses UNet1DModel with a block_out size of 64 to create random and conditional vectors, combined with a diffusion process (e.g., 1000 diffusion steps) to generate rich data that accurately reflects the complex features of real data, thereby supporting the detection of sophisticated attack patterns hidden in normal data.

Second, data discrimination and optimization: The Diffusion model integrates real and generated data to train a discriminator model, with continuous optimization via a scheduler like DDPMScheduler, ensuring that the generated data closely resembles real data and enhancing classification capability.

Third, creation of a balanced dataset: The Diffusion model combines generated and real data to build a balanced dataset, reducing imbalance and strengthening the network of relationships between flows. This is particularly effective when Window = 10 is used to exploit the broad context from 10 flows before and after the APT center.

In comparison with VAE-LSTM, that model only achieved an accuracy of 97.00%, precision of 84.00%, recall of 87.00%, and an F1-score of 85.00%. Its significant drawbacks include:

First, its limited data generation capability. VAE relies on a simple Gaussian distribution and only has 2 layers, leading to less diverse synthetic data that cannot fully reflect complex features or long-term relationships in the real data, which reduces the performance in detecting sophisticated attack patterns.

Second, it has no effective noise removal mechanism. VAE-LSTM does not integrate a diffusion process like our model, which diminishes its ability to distinguish between generated and real data, especially in imbalanced datasets.

In comparison with KNN-SMOTE-LSTM, that model achieved an accuracy of 96.00, precision of 72.00, recall of 93.00, and an F1-score of 81.00%. Its serious drawbacks include:

Table 4 APT attack detection results with synthetic data generation

Model			Evaluation of IP			
			Acc	Pre	Rec	F1
ACDF-mLSTM			**0.9980**	**0.9850**	**0.9900**	**0.9875**
Window (Flows before/after APT center)	Diffusion block_out	mLSTM layers				
10	**64**	**2**				
VAE-LSTM[19]			0.97	0.84	0.87	0.85
VAE layers		LSTM layers				
2		3				
KNN-SMOTE-LSTM[17]			0.96	0.72	0.93	0.81
MLP layers	SMOTE	GCN layers				
2		3				
BiLSTM-attention-dropout[18]			0.9761	0.9761	0.9761	0.9753
BiLSTM layers	Attention layers	Dropout				
2	3					

Bold indicates the superior performance of the proposed method

First, a simple and inefficient data balancing method. SMOTE generates synthetic data through linear interpolation between neighboring points with the support of 2 MLP layers, but it fails to capture multidimensional relationships or latent features. This leads to noisy generated data and a precision of only 72.00%, which in turn results in a low F1-score.

Second, a lack of continuous optimization capability. KNN-SMOTE-LSTM does not have a diffusion or attention mechanism like the Diffusion model and mLSTM, making the model unable to adjust when faced with imbalanced data, leading to poor overall performance.

Third, limitations in processing long-term context. With 3 GCN layers, it focuses on graph structure but does not exploit sequential context like the Window = 10 in ACDF-mLSTM, which reduces its ability to analyze relationships between flows, especially with complex APT data.

Figure 5 illustrates the change in the correlation relationships between data attributes before and after applying the Diffusion model. Before using Diffusion, the attributes exhibited non-uniform correlation levels, with some attribute pairs having strong links while others had negligible correlation. After applying Diffusion, the correlation chart shows a more uniform distribution among attributes, reflecting the model's effectiveness in adjusting these relationships. This indicates that the Diffusion model not only generates synthetic data to balance sample counts but also enhances data diversity and representation by refining features, particularly through the UNet1DModel with an optimal block_out size (such as 64). This improvement supports the ACDF-mLSTM model in capturing complex relationships within the data, thereby enhancing its ability to detect and classify APT threats. The critical role of the Diffusion model is affirmed as it optimizes the quality of the input data, ensuring not only numerical balance but also informational depth.

4.4.3 Experimental results for scenario 3

As previously stated, the research question for this scenario is to explain and evaluate the effectiveness of the ACDF-mLSTM model in aggregating and highlighting the important information of APT IPs. In Table 5 below, the paper presents the experimental results comparing the ACDF-mLSTM model with several other models and approaches.

Based on the results in Table 5, the ACDF-mLSTM model achieves outstanding performance in detecting APT attacks when handling imbalanced data, thanks to the prominent role of mLSTM. With an accuracy of 99.80, precision of 98.50, recall of 99.00, and an F1-score of 98.75% in its optimal configuration (Window = 10, mLSTM Layers = 2), ACDF-mLSTM significantly outperforms the BiLSTM-Attention, GCN-BiLSTM, and MIG models. This is attributed to the following factors: mLSTM, with its 2 layers, 4-head attention mechanism, Conv1D, Learnable Skip Connection, and Group Normalization, processes data in parallel through two branches. One branch uses Conv1D and attention to aggregate local information, while the other uses an MLP with a Swish function as an output gate to ensure control over the information flow and effectively capture long-term context. The use of Window = 10 to exploit 10 flows before and after the APT center enhances the ability to analyze complex relationships, thereby improving classification performance.

Furthermore, a comparison of the ACDF-mLSTM model with other models reveals the following:

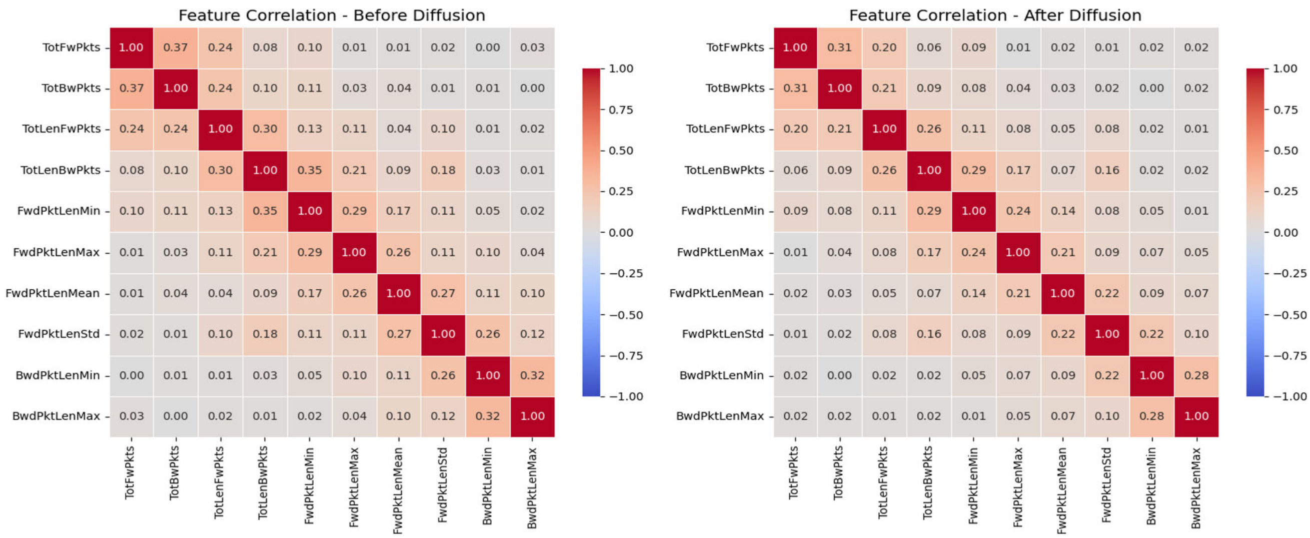


Fig. 5 Feature correlation chart before and after applying the diffusion model

Table 5 APT attack detection results for feature aggregation and extraction

Model			Evaluation of IP			
			Acc	Pre	Rec	F1
ACDF-mLSTM			**0.9980**	**0.9850**	**0.9900**	**0.9875**
Window (Flows before/after APT center)	Diffusion block_out	mLSTM layers				
10	**64**	**2**				
BiLSTM-attention [19]			0.97	0.86	0.86	0.86
BiLSTM layers		Attention layers				
2		3				
GCN-BiLSTM[31]			0.97	0.89	0.90	0.89
GCN layers		BiLSTM layers				
3		2				
MIG[32]			0.9732	0.9731	0.9742	0.9740
MLP layers	Coefficient β of inference	GCN layers				
3	0.7	2				

Bold indicates the superior performance of the proposed method

Comparison with BiLSTM-Attention: This model only achieved an accuracy of 97.00, precision of 86.00, recall of 86.00, and an F1-score of 86.00% with 2 BiLSTM layers and 3 Attention layers. Its drawbacks include: first, a limited attention mechanism, where 3 Attention layers are not as optimized as mLSTM's 4-head attention and fail to effectively exploit complex relationships, leading to low recall and precision. Second, a lack of parallel processing mechanisms, as BiLSTM-Attention does not integrate Conv1D or Learnable Skip Connection, which reduces its ability to aggregate local information. It also lacks Group Normalization, making the model unstable when handling imbalanced data and unable to achieve the same sensitivity as mLSTM.

Comparison with GCN-BiLSTM: This model achieved an accuracy of 97.00, precision of 89.00, recall of 90.00, and an F1-score of 89.00% with 3 GCN layers and 2 BiLSTM layers. The model is not truly effective for several reasons: first, its focus on graph structure, where 3 GCN layers prioritize local relationships but do not handle long-term sequential context as well as mLSTM with Window = 10. Second, a weaker BiLSTM component, where 2 BiLSTM layers do not integrate advanced attention or parallel mechanisms like mLSTM, leading to lower classification performance. It also lacks an information flow control mechanism, such as the output gate in mLSTM, which reduces its ability to optimize the output.

Comparison with MIG: This model achieved an accuracy of 97.32, precision of 97.31, recall of 97.42, and an F1-score of 97.40% with 3 MLP layers, an inference coefficient $\beta = 0.7$, and 2 GCN layers. Our analysis of the MIG model revealed several limitations: (i) dependency on simple inference, where with $\beta = 0.7$, MIG focuses only on the weights of APT flows without exploiting the broad context provided by mLSTM's Window = 10; (ii) lack of an advanced attention mechanism, as its 2 GCN and 3 MLP layers cannot compete with mLSTM's 4-head attention in processing complex relationships; and (iii) no parallel processing or Group Normalization, which reduces its stability and effectiveness when handling complex data.

4.4.4 Experimental results for scenario 4

Due to the nature of these three datasets, which consist of individual flows rather than frames and lack IP information, our model's context length was adjusted to 1 to suit the data's characteristics. Below are the evaluation metrics of our model on these new datasets compared to other models:

Based on (Table 6), which compares the performance of APT attack detection models on the BoT, ToN, and IDS2018 datasets, the ACDF-mLSTM model stands out with superior performance compared to other models such as E-GraphSAGE, Anomal-E, E-ResGAT, and FeCoGraph. Specifically, on the BoT dataset, ACDF-mLSTM achieved an Accuracy of 99.82, Precision of 83.94, Recall of 98.33, and F1-Score of 89.54%, surpassing E-GraphSAGE (Accuracy 99.65, Precision 97.37, Recall 40.12%), Anomal-E (Precision 66.58%), and FeCoGraph (Recall 51.05 despite an Accuracy of 99.97%). On the ToN dataset, the model recorded an Accuracy of 98.62, Precision of 98.53, Recall of 97.99, and F1-Score of 98.26%, outperforming E-GraphSAGE (78.47%), Anomal-E (95.77%), E-ResGAT (93.84%), and FeCoGraph (96.90%). Notably on the IDS2018 dataset, ACDF-mLSTM achieved an Accuracy of 99.46, Precision of 99.65, Recall of 97.78, and F1-Score of 98.69%, significantly higher than E-GraphSAGE (93.20%), Anomal-E (91.57%), E-ResGAT (98.45%), and FeCoGraph (99.63%).

The superior performance of ACDF-mLSTM can be explained by the unique combination of its core components. First, AC removes noise and focuses on important features, optimizing resources and enhancing the ability to identify sophisticated attack patterns. Second, the Diffusion Model creates diverse and balanced attack data vectors, effectively addressing the data imbalance problem, thereby improving accuracy and sensitivity. Third, mLSTM, with its ability to process sequential data, store information in matrix form, and integrate an Attention mechanism, allows the model to capture long-term relationships and focus on critical data, optimizing contextual analysis.

In comparison, E-GraphSAGE, despite high Accuracy (99.65% on BoT), had low Recall (40.12%), reflecting its limitations in flexible context processing due to its reliance on graph representations. Anomal-E, with an Accuracy of 99.87% on BoT but Precision of only 66.58% in some configurations, shows shortcomings in handling imbalanced data, leading to uneven performance. E-ResGAT, despite an Accuracy of 98.45% on IDS2018, had a lower Recall (95.30%), indicating difficulties in detecting minority attack patterns due to the lack of a strong data balancing mechanism. FeCoGraph, while achieving high Accuracy (99.97% on BoT), had a very low Recall (51.05%), as it focuses on feature fusion without effectively handling noise or imbalance.

Figure 6 shows the Confusion Matrix results for the three datasets. The matrices display the True Positives, False Positives, False Negatives, and True Negatives for the model on each dataset. The first matrix corresponds to the NF-ToN-IoT-v2 dataset, the second to the NF-BoT-IoT-v2 dataset, and the third to the NF-CSE-CIC-IDS2018-v2 dataset.

4.5 Discussion

Based on diverse evaluation scenarios, we have demonstrated the superiority of the ACDF-mLSTM model over existing methods. This advantage stems from two primary factors:

First, the data balancing solution with ACDF. ACDF allows the model to flexibly adjust to the different characteristics of network traffic while eliminating noise and prioritizing important features through its data preprocessing stage. By exploiting the sequential relationships between data flows, particularly by considering preceding and succeeding data points, ACDF creates a solid foundation for identifying sophisticated anomalies, thereby enhancing the accuracy of APT detection. Furthermore, ACDF applies a Diffusion Model with UNet1DModel, using parameters such as block_out size (e.g., 64) and the number of diffusion steps (e.g., 1000), to generate new synthetic attack data. This method not only simulates realistic attack tactics but also balances the dataset by integrating real data with generated data, helping to minimize imbalance and provide a rich dataset. This supports the model in learning from various attack scenarios, thereby improving its ability to detect previously unseen patterns.

Second, the optimization of feature extraction and aggregation with mLSTM. The mLSTM component, with its 2 layers, 4-head attention mechanism, Conv1D, Learnable Skip Connection, and Group Normalization, excels at performing two tasks: extracting information from network traffic flows and highlighting important features before classification. Thanks to parallel processing through two branches—one using Conv1D and attention to focus on local information, and the other using an MLP with a Swish function as an

Table 6 Experimental results evaluating the proposed model on other datasets

Dataset	Method	Accuracy	Precision	Recall	F1_Score
BoT	E-GraphSAGE [33]	99.65	**99.82**	51.05	51.97
	Anomal-E [34]	99.87	97.37	83.94	89.54
	E-ResGAT [35]	99.74	66.58	40.12	44.57
	FeCoGraph [36]	99.89	96.92	86.61	91.12
	ACDF-mLSTM	**99.97**	97.97	**98.70**	**98.33**
ToN	E-GraphSAGE [33]	78.47	79.80	82.15	78.25
	Anomal-E [34]	95.77	95.88	94.89	95.36
	E-ResGAT [35]	93.84	94.23	92.46	93.23
	FeCoGraph [36]	96.90	96.81	96.44	96.62
	ACDF-mLSTM	**98.62**	**98.53**	**97.99**	**98.26**
IDS2018	E-GraphSAGE [33]	93.20	92.49	73.40	79.36
	Anomal-E [34]	91.57	83.03	72.22	76.17
	E-ResGAT [35]	98.45	97.34	95.30	96.29
	FeCoGraph [36]	**99.63**	99.38	**98.85**	**99.11**
	ACDF-mLSTM	99.46	**99.65**	97.78	98.69

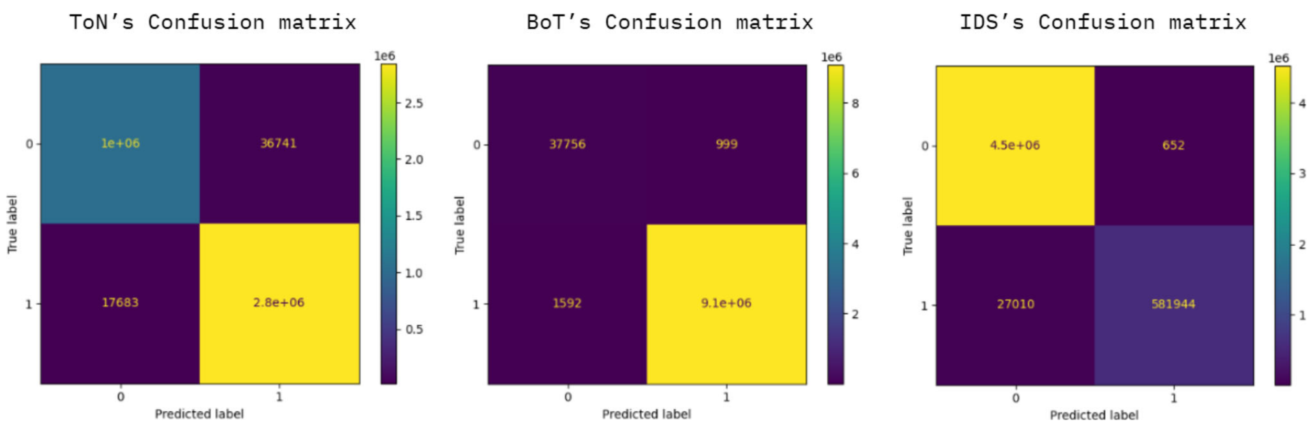


Fig. 6 Confusion matrices of the model on the three datasets: NF-ToN-IoT, NF-BoT-IoT, and NF-CSE-CIC-IDS2018

output gate—mLSTM efficiently exploits sequential relationships and complex dependencies, enhancing analytical capabilities and leading to accurate decisions.

The experimental results have confirmed the efficacy of ACDF-mLSTM, with an average improvement of 2 to 12% across all evaluation metrics, outperforming other methods on various datasets and scenarios. The combination of ACDF's advanced data generation technique and mLSTM's feature extraction capability not only enhances accuracy but also ensures adaptability in complex data situations. However, the model still faces certain limitations:

In some specific cases, such as when processing network data with complex or frequently changing structures, ACDF may require additional time to adjust and fully capture the hidden relationships between data points. For instance, in a large enterprise network with tens of thousands of devices—including computers, servers, and IoT devices—using diverse protocols like HTTP, FTP, or custom protocols, the data may be non-uniform due to noise from transmission errors, hardware failures, or abnormal yet legitimate activities. This could slightly affect performance, especially during the initial phase when new data is introduced. The cause stems from ACDF's dependency on input data quality; when data is fragmented or lacks clear patterns, the data generation process may require further fine-tuning. This is a natural aspect of real-world network environments, which are often volatile and unpredictable.

On the mLSTM side, despite its impressive performance in processing sequential data, the model may need more time to adapt to new attack scenarios, such as zero-day attacks with unique characteristics. For example, an attack using a new data encryption method or exploiting an unrecorded vulnerability might require mLSTM to be supplemented with additional training data to adjust its attention weights. This reflects mLSTM's learning process, which is based on initial data, and fine-tuning parameters like the number of attention heads or Conv1D filters may need careful consideration to maintain effectiveness without causing imbalance. This presents an opportunity for future model improvement when facing increasingly creative network threats.

5 Conclusion and future work

The detection of APT attacks specifically, and network attacks in general, is and will always be an urgent issue for monitoring and intrusion detection systems. In this paper, with the objective of enhancing the ability to accurately detect APT IPs and reducing false alarm rates, we have successfully proposed a new ensemble learning model based on the combination of data balancing and feature extraction/aggregation techniques. In the experimental section, to demonstrate the effectiveness of the proposed model, the paper conducted multiple scenarios on various datasets. The experimental scenarios show that the two major problems posed in this study have been significantly improved. Specifically, with a more accurate and superior data generation capability, the data generation technique proposed in the paper was more effective than some other traditional methods like SMOTE, Dropout, and VAE. Similarly, by better understanding the data's context and content, the proposed model for feature extraction and aggregation was more effective than hybrid deep learning models, deep graph networks, and others. These results mark a major and novel step forward in the detection of APT IPs in particular, and network attacks in general, based on ensemble learning techniques.

In the future, to continue improving the model's accuracy, we believe several issues can be addressed as follows [37]:

First, to address the data imbalance problem, new data generation solutions from fields such as image recognition or adversarial machine learning could be used. These techniques would support the creation of new datasets without relying on the original data, which would help create rich datasets containing diverse attack characteristics. Alternatively, one could consider applying advanced new deep learning models like MLP-Swish or xLSTM to better understand data structure and content, thereby achieving better feature aggregation. This is a new approach that seeks to use more optimal methods rather than just trying to balance the data through generation. This approach would make the model less cumbersome and complex [38].

Second, regarding the task of [39] feature extraction and aggregation. Based on the experimental results, it is clear that the direction proposed in this paper is sound and reasonable. Therefore, to further improve efficiency, future research could focus on optimizing some of this model's parameters by stacking nLSTM blocks, allowing the model to make an optimal choice rather than being limited to a single one [40]. In addition, some advanced Transformer models currently available, such as Rotary Transformer, Mamba, and DeepSeek, would all be good options to explore. However, their application would require architectural improvements to the layers and blocks to ensure suitability and to optimize computational speed.

Author contributions Cho Do Xuan raised the idea, initialized the project, and designed the experiments; Long and Duc carried out the experiments under the supervision of Cho Do Xuan. Both authors analyzed the data and results. Cho Do Xuan wrote the paper.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interests.

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